Problem 5

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Problem 1 *Determine the following information about the given function and then graph the function:*

$$f(x) = x\ln(x)$$

Domain
x, y-intercepts
Symmetry
Intervals of increasing/decreasing
Maxima/Minima (including absolute)
Intervals of concavity
Inflection points
Given $\lim_{x\to 0^+} f(x) = 0$, sketch a graph f.

Explanation. (a) Domain: $(0, \infty)$

(b) x, y-intercepts:
x-interpects occur when $f(x) = x\ln(x) = 0$. We would think this occurs when either $x = 0$ or $ln(x) = 0$. However, $x = 0$ is not in the domain. We only have $ln(x) = 0$ which is at $x = 1$. y-intercepts: None given the domain.

(c) Symmetry: Given the domain of the function, it does not make sense to check for even/odd.

(d) Intervals of increasing/decreasing:

$$f'(x) = \ln(x) + 1$$

Find: $f'(x) = \ln(x) + 1 = 0 \implies \ln(x) = -1 \implies x = e^{-1}$

Since f' possibly changes its sign at critical points only, the sign of f' is constant on intervals $(0, e^{-1})$ and (e^{-1}, ∞). Since $f'(1) = 1 > 0$, and $f'(e^{-3}) = -3 + 1 = -2 < 0$, it follows that f' is negative on $(0, e^{-1})$ and positive on (e^{-1}, ∞).

Therefore, f is decreasing on $(0, e^{-1})$ and f is increasing on (e^{-1}, ∞).

(e) Maxima/Minima
Since the function is decreasing on $(0, e^{-1})$ and f is increasing on (e^{-1}, ∞)., the absolute minimum value of f is attained at $x = e^{-1}$. The absolute minimum value of $f(x) = f(e^{-1}) = e^{-1}\ln e^{-1} = -e^{-1} = -\dfrac{1}{e}$

The absolute maximum value of f does not exist, since the only critical point is a minimum and $\lim_{x\to\infty} f(x) = +\infty$

(f) Intervals of concavity and inflection points: $f'(x) = \ln(x)+1 \implies f''(x) = \frac{1}{x} > 0$ on the domain of f. Concave up on its domain and there are no inflection points.

(g) Given $\lim_{x\to 0^+} f(x) = 0$, sketch a graph f: